PROJECT REPORT

Task Title: Network Traffic Investigation & Attack Identification
Tools Used: Kali Linux, Wireshark, Tshark
Evidence: Screenshots from the investigation

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1. Objective

The objective of this task was to analyze the provided `Task 4.pcap` file and identify unusual network activity, suspicious communication patterns, possible attack behavior, and the security impact of the observed traffic.

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2. Work Performed

I opened and analyzed the PCAP file using **Wireshark and Tshark in Kali Linux**.

The initial analysis showed:

- Total packets: **69,833**
- Capture duration: approximately **30 minutes**
- Main protocols observed: **TCP, SSH, UDP, DNS and ARP** Main internal hosts observed:
  - `192.168.0.1`
  - `192.168.0.20`
  - `192.168.0.21`

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3. Network Traffic Analysis

During the investigation, I checked the communication between the observed hosts.

A notable traffic pattern was found between:

`192.168.0.20 → 192.168.0.1`

The source host made repeated TCP connection attempts to multiple destination ports.

Some of the observed ports included:

```
21
22
23
25
53
80
113
389
443
636
1099
1723
3389
```

The target host also generated TCP reset responses for the examined connection attempts.

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4. Suspicious Activity

The repeated TCP connection attempts to multiple ports on the same target were identified as suspicious network behavior.

The pattern is **consistent with TCP port/service scanning**.

Source Host

```
192.168.0.20
```

Target Host

```
192.168.0.1
```

Protocol

TCP

Activity

Multiple connection attempts to different destination ports.

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5. Security Assessment

The observed activity may indicate network reconnaissance or service enumeration.

If the activity was unauthorized, it could allow a user or attacker to identify services available on the target system.

However, the PCAP alone does not confirm:

- Successful exploitation
- System compromise
- Malware execution
- Data theft

Therefore, the assessment is limited to the network behavior directly observed in the PCAP.

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6. MITRE ATT&CK Mapping

The observed activity can be associated with:

Technique: Network Service Scanning
Technique ID: T1046

Reason

Multiple TCP connection attempts were observed against different ports on the target host. This behavior is consistent with identifying available network services.

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7. Conclusion

The `Task 4.pcap` file was successfully analyzed using Kali Linux, Wireshark and Tshark.

The investigation identified repeated TCP connection attempts from `192.168.0.20` toward multiple ports on `192.168.0.1`.

Based on the available packet evidence, the activity is consistent with **TCP port/service scanning**.

No successful compromise was confirmed from the available PCAP evidence.

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8. Evidence Attached

The following screenshots are attached as supporting evidence:

- PCAP information and packet count
- Protocol statistics
- Host communication analysis
- TCP SYN/connection attempts
- Destination port activity
- TCP reset responses
- Relevant Wireshark/Tshark analysis

Attachment: CSA-004 Investigation Report with supporting screenshots.

DAILY WORK REPORT

Subject: Daily Work Report – Amal Sam (21-09-2026)

Task Completed

- **Phase 1 – Network Overview** ○ Analyzed the provided PCAP file.
  - Identified packet count, capture duration, hosts and protocols.
- **Phase 2 – Host Communication Analysis** ○ Analyzed source and destination IP addresses.
  - Identified the major communicating hosts.
- **Phase 3 – Traffic Pattern Investigation** ○ Investigated repeated TCP connections.
  - Checked destination ports and connection responses.
  - Identified a possible port scanning pattern.
- **Phase 4 – Attack Identification** ○ Investigated the suspicious traffic between `192.168.0.20` and `192.168.0.1`. ○ Identified possible TCP service/port scanning.

On-Going

- **Phase 5 – Timeline Reconstruction** ○ Organizing timestamps and important network events.
- **Phase 6 – Threat Assessment** ○ Preparing the security impact assessment.
- **Phase 7 – MITRE ATT&CK Mapping**
  - Documenting the relevant technique and justification.

Pending

- **Phase 1 – Final Report Submission**
  - Complete the final report.
  - Attach investigation screenshots.
  - Submit the CSA-004 task.